

# Karthik Viswanathan

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I am currently pursuing a PhD at the University of Amsterdam, focusing on **interpretability** of large language models through **geometric analysis** of their internal representations. My interdisciplinary background began with **competitive programming**, expanded through developing machine learning models as a **surveillance analyst** at Goldman Sachs, and was further enriched during my master's studies in **theoretical physics**. This diverse background has enabled me to learn new concepts and adapt to new domains relatively easily, allowing me to contribute to both implementation and theoretical aspects of scientific research.

## Education

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**University of Amsterdam** **Amsterdam, Netherlands**  
*PhD in Physics, Expected: December 31, 2025* *2021–2025*  
**Thesis:** From Language Models to Cosmic Structures: A Geometric Perspective  
**Advisor:** Dr. Jan Pieter van der Schaar

**University of Amsterdam** **Amsterdam, Netherlands**  
*MSc in Physics and Astronomy (Theoretical Physics), cum laude* *2019–2021*  
**Thesis:** Exploring the Spectral Theory/Topological Strings duality  
**Advisor:** Dr. Marcel Vonk  
**GPA:** 8.7/10

**Indian Institute of Technology Madras** **Chennai, India**  
*B.Tech in Engineering Physics* *2013–2017*  
**Thesis:** Real Space Renormalization and Applications to Machine Learning  
**Advisor:** Dr. Ashwin Joy  
**GPA:** 8.29/10

## Research Experience

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**Probing Geometry of Next Token Prediction Using Cumulant Expansion of Softmax Entropy**  
**Karthik Viswanathan**, Sang Eon Park  
*ICML 2025: Workshop on High-dimensional Learning Dynamics*

**Persistent Topological Features in Large Language Models**  
Yuri Gardinazzi\*, **Karthik Viswanathan\***, Giada Panerai, Alessio Ansuini, Alberto Cazzaniga, Matteo Biagetti  
*Proceedings of the 42nd International Conference on Machine Learning* *\*Equal contribution*

**The Geometry of Tokens in Internal Representations of Large Language Models**  
**Karthik Viswanathan**, Yuri Gardinazzi, Giada Panerai, Alberto Cazzaniga, Matteo Biagetti  
*Under review*

**Cosmology with persistent homology: A Fisher forecast**  
Jacky Yip, Matteo Biagetti, Alex Cole, **Karthik Viswanathan**, Gary Shiu  
*Journal of Cosmology and Astroparticle Physics (JCAP), Vol. 2024, Issue 09*

**Workshop: Interpretability in LLMs using Geometric and Statistical methods**  
**Karthik Viswanathan**, Jan Pieter van der Schaar  
*Organized a two-day workshop with around 40 participants from Amsterdam and AREA Science Park (Trieste).*

## Professional Experience

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| <b>Goldman Sachs</b><br><i>Surveillance Analyst</i><br>Developed ML models for anomaly detection in financial data<br><b>Manager:</b> Prof. Howard Karloff    | <b>Bangalore, India</b><br>2017–2019 |
| <b>Goldman Sachs</b><br><i>Summer Intern</i><br>Implemented fast Personalized PageRank in a MapReduce framework<br><b>Mentor:</b> Dr. Koushik Balasubramanian | <b>Bangalore, India</b><br>2016      |

## Awards & Distinctions

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| <b>Sander Bais Prize for Academic Merit</b><br><i>Institute for Theoretical Physics Amsterdam</i><br>For exceptional academic performance in the master's program   | <b>Amsterdam, Netherlands</b><br>2020 |
| <b>Honorable Mention</b><br><i>ACM ICPC World Finals</i><br>Represented India in the international collegiate programming contest.                                  | <b>Marrakech, Morocco</b><br>2015     |
| <b>Participant</b><br><i>International Olympiad in Informatics Training Camp</i><br>Selected after ranking in top 25 in the Indian National Olympiad in Informatics | <b>Bangalore, India</b><br>2013       |

## Organization & Mentoring

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| <b>AREA Science Park, Trieste</b><br><i>Master Student Supervisor</i><br><b>Student:</b> Giada Panerai<br><b>Thesis:</b> Zig-Zag Persistence in Neural Networks Representations                                        | 2024 |
| <b>University of Amsterdam</b><br><i>Master Student Mentor</i><br><b>Student:</b> Sibilla Bouche<br><b>Thesis:</b> The persistence of non-Gaussian features: a Neural Ratio Estimation approach                        | 2023 |
| <b>University of Amsterdam</b><br><i>Master Student Supervisor</i><br><b>Student:</b> Sliem el Ela<br><b>Thesis:</b> From Primordials to Persistence: A Dual Exploration of Multiparameter Topology and Cosmic Origins | 2023 |
| <b>University of Amsterdam</b><br><i>Journal Club Organizer</i><br><b>Cosmology Journal Club</b>                                                                                                                       | 2022 |

## Technical Skills

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**Programming:** PyTorch, TensorFlow, Hadoop, MapReduce, Mathematica  
**HPC Resources:** LUMI consortium, Snellius supercomputer

## Teaching

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**University of Amsterdam**

*Teaching Assistant*

**Courses:** Topological Data Analysis: a Physics Perspective, Advanced Cosmology, ML for Physics and Astronomy, QFT in Curved Spacetimes, General Relativity

**MSc Physics Program**

*2020–2024*

## Research Visits

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**2024: AREA Science Park**, Trieste, Italy

September 16 – October 4

**2024: AREA Science Park**, Trieste, Italy

February 15 – April 27

## References

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**Dr. Jan Pieter van der Schaar**

University of Amsterdam

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**Dr. Matteo Biagetti**

AREA Science Park, Trieste

[matteo.biagetti@areasciencepark.it](mailto:matteo.biagetti@areasciencepark.it)

**Dr. Magnus Botnan**

Vrije Universiteit, Amsterdam

[M.B.Botnan@vu.nl](mailto:M.B.Botnan@vu.nl)

**Dr. Alberto Cazzaniga**

AREA Science Park, Trieste

[alberto.cazzaniga@areasciencepark.it](mailto:alberto.cazzaniga@areasciencepark.it)